

Electronics Workshop

The meter, switches, simple circuits, and the Electrical Symbols handout.

One three-hour session. Measure and read — not yet wiring the robot.

ORDER / CONFIRM

Simple Circuits Lab Kits in hand (ordered in 1.3). 9 V batteries stocked.

T-1 WEEK

Assign MATE pre-watch. Print the Electrical Symbols handout, one per student.

T-1 DAY

Stage meters + leads, lab kits, components, the symbols handouts.

T-0

Safety brief. Meters out.

ORDER AHEAD · HAVE ON HAND

- ☐ **CONFIRM NOW** SeaMATE Simple Circuits Lab Kit — one per group, in hand (ordered in Class 1.3).
- ☐ Multimeters (one per pair) + test leads — owned.
- ☐ 9 V batteries for the lab kits.
- ☐ Electrical Symbols handout — free to print, one per student. **Keep forever.**
- ☐ Spare LEDs, resistors, switches, and jumper wire for extra practice.

PRE-WATCH HOMEWORK · ASSIGN BEFORE CLASS

- ☐ MATE: *Multimeter Basics*
- ☐ MATE: *How Switches Work*
- ☐ MATE: *Simple Circuits*

Let MATE teach the concepts; protect meeting time for measuring and building.

STAGE IN THE ROOM

- ☐ Multimeters (one per pair) + test leads
- ☐ Simple Circuits Lab Kit per group (+ a 9 V battery each)
- ☐ Loose components for the build-and-measure rep
- ☐ Printed Electrical Symbols handout, three-hole punched
- ☐ Project notebook for schematic-reading practice

PREP OWNERS

- ☐ **CFO + ADVISOR** — confirm lab kits + 9 V batteries on hand
- ☐ **ADVISOR** — assign MATE pre-watch; print symbols handout (T-1 wk)

SAY WATT ROBOTICS · ROV TEAM

SET UP · SAFETY

- ☐ Bench space per pair; no shorting battery terminals
- ☐ Irons in stands if a lead needs tinning
- ☐ Meters set to the right range before connecting

CUE UP · IN-ROOM REFERENCE

- ☐ MATE: *TriggerFish Controller Simplified* (preview of the control box)
- ☐ The Electrical Symbols handout on screen
- ☐ Instructor Notes — the Ohm's-law and open/short grounding

STUDENTS BRING

- ☐ Watched the MATE pre-watch
- ☐ Their notebook for schematic practice

CONFIRM · GATING

- ☐ Control-box parts + soldering stations staged for Class 1.5 (a long build)
- ☐ 12 V battery + charger in hand before the wiring lessons
- ☐ Repo ready for wiring notes (Software Lead)

- ☐ **ADVISOR** — stage meters, kits, components (T-1 day)

- ☐ **CEO + ADVISOR** — run safety brief; stage 1.5 control-box parts

CLASS 1.4 — ELECTRONICS WORKSHOP · PREP SHEET